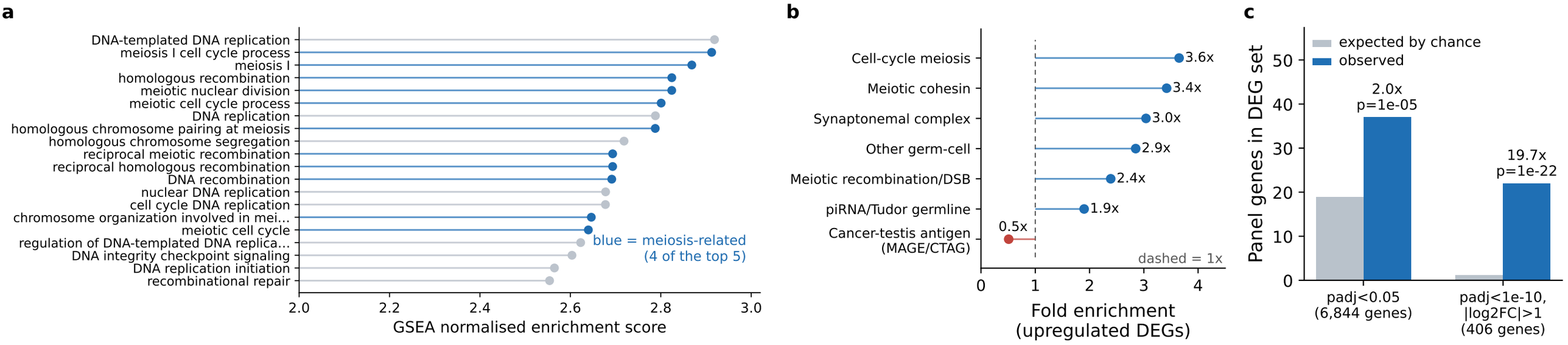

a
DNA-templated DNA replication
meiosis I cell cycle process
meiosis I
homologous recombination
meiotic nuclear division
meiotic cell cycle process
DNA replication
homologous chromosome pairing at meiosis
homologous chromosome segregation
reciprocal meiotic recombination
reciprocal homologous recombination
DNA recombination
nuclear DNA replication
cell cycle DNA replication
chromosome organization involved in mei...
meiotic cell cycle
regulation of DNA-templated DNA replica...
DNA integrity checkpoint signaling
DNA replication initiation
recombinational repair
blue = meiosis-related
(4 of the top 5)
2.0
2.2
2.4
2.6
2.8
3.0
GSEA normalised enrichment score
b
Cell-cycle meiosis
Meiotic cohesin
Synaptonemal complex
Other germ-cell
Meiotic recombination/DSB
piRNA/Tudor germline
Cancer-testis antigen
(MAGE/CTAG)
3.6x
3.4x
3.0x
2.9x
2.4x
1.9x
0.5x
dashed = 1x
0
1
2
3
4
Fold enrichment
(upregulated DEGs)
c
Panel genes in DEG set
expected by chance
observed
2.0x
p=1e-05
19.7x
p=1e-22
0
10
20
30
40
50
padj<0.05
(6,844 genes)
padj<1e-10,
|log2FC|>1
(406 genes)